

Production NOK Increase

RCA Training Case

Simulated / Training Case — not real workplace experience

Case type	Simulated / Training
Methods	RCA · Fishbone (6M) · Hypothesis Testing · 5 Why
Purpose	Practical application of structured root-cause analysis

1. Problem Statement

Problem: Production NOK increased from 2% to 8%. The issue was observed mainly on Machine B and was more frequent during the night shift.

Objective: identify and validate potential causes instead of assuming a single cause.

2. Initial Observation

The initial observation pointed toward Machine B and the night shift. Potential differences in operator behaviour, machine conditions, material, process method, measurement and environmental conditions were considered.

3. Fishbone / 6M — Potential Causes

Category	Potential causes
Man	Incorrect execution of the standard; fatigue/reduced concentration; differences between operators.
Machine	Incorrect machine parameters; controller-related errors; unstable process settings.
Material	Different material behaviour; batch/lot differences; storage conditions; supplier/source differences; material age.
Method	Process not performed according to the work instruction; skipped process step; difference between documented and actual method.
Measurement	Incorrect measurement method, instrument or execution; differences between shifts/operators.
Environment	Ambient temperature and humidity; possible night/day environmental differences.

4. Hypothesis Development and Validation

Potential causes were converted into testable hypotheses. The aim was to keep other relevant conditions as consistent as possible and compare the resulting defect rate.

Hypothesis	Result	Assessment
Temperature	NOK 8% → 7.8% after changing temperature.	Rejected as significant cause.
Operator	NOK 8% → 8.1% after changing operator.	Rejected as primary cause.
Measurement	Two operators produced practically identical results on the same element.	Rejected as primary cause.
Machine parameters	Restoring standard parameters reduced NOK 8% → 3%.	Strongly supported.
Material	One batch showed 27% NOK; other batches were around 3–5%.	Strong suspect; further validation needed.
Method	A skipped instruction step was found; restoring it reduced NOK 8% → 3.2%.	Strongly supported.

5. 5 Why Analysis

Problem: Machine B NOK increased to 8%, mainly during the night shift.

Why 1? The instruction step was being skipped.

→ The instruction was not correctly implemented on the night shift.

Why 2? Why was it not correctly implemented?

→ The person responsible did not ensure that the change was implemented.

Why 3? Why was implementation not ensured?

→ The responsible person was absent/on leave and responsibility was not properly transferred.

At this point the analysis moves from an individual action toward a system-level weakness: continuity of responsibility for implementing instruction changes was not ensured.

6. Root Cause

Root cause: Lack of a reliable mechanism ensuring continuity of responsibility for implementing work-instruction changes during the absence of the responsible person.

7. Corrective Action

Implement the specific work-instruction change on the night shift and verify that the affected process is being performed according to the current standard.

8. Preventive Action

Establish a defined backup/deputy responsibility for instruction changes and a confirmation mechanism showing that revised instructions have been communicated and implemented across all relevant shifts.

9. Key Learning

RCA should move from observations and assumptions toward testable hypotheses, evidence-based validation and systemic root causes. A person making an error may be part of the causal chain, but the investigation should also ask why the system allowed the condition to occur.

The 6M structure organized potential causes, hypothesis testing separated weaker explanations from stronger ones, and 5 Why helped move from an immediate process failure toward a controllable system-level cause.

10. Case Classification & Limitations

Classification: Simulated / Training Case.

The scenario, percentages and test results are simulated for learning purposes. This case does not represent a real workplace investigation and must not be presented as professional work experience.